October 25, 2026

PERSONAL STATEMENT OF VIVEK MISHRA IN SUPPORT OF PETITION FOR ALIEN OF EXTRAORDINARY ABILITY (EB-1A)

Dear Immigration Officer,

I, Vivek Mishra, respectfully submit this personal statement in support of my I-140 Petition for classification as an Alien of Extraordinary Ability (EB-1A). I am an AWS Cloud Consultant, software engineer, and researcher specializing in the intersection of scalable backend systems, Artificial Intelligence (AI), and high-availability financial technology (FinTech). I currently serve The Vanguard Group in Malvern, Pennsylvania, through Randstad Digital, and from October 2021 to May 2026 I served as a backend engineer at JPMorgan Chase & Co.

A dual expertise defines my career. First, I am an engineer who designs and operates mission-critical cloud and FinTech infrastructure for two of the most distinguished financial institutions in the United States, JPMorgan Chase and The Vanguard Group. Second, I am an applied researcher whose work on AI and optimization in public transportation, funded by the U.S. Department of Transportation, has been cited by researchers in Florida, Italy, Iran, and Colombia.

My achievements have been recognized by peers at the top of my field, in both industry and academia, confirming my sustained national and international acclaim. This recognition is documented in the supporting letters of a Director of Software Engineering and a Vice President of Engineering at JPMorgan Chase, university professors and researchers in transportation engineering, and senior AI practitioners, all of whom have independently assessed my work.

I. Proposed Endeavor and Plans to Continue My Work in the United States

I intend to continue working in the United States in the same field in which I have established my expertise: building secure, scalable, and intelligent cloud and backend systems for financial services, and applying AI and optimization to public infrastructure. I describe my plans below in three stages.

Immediate plans (now). I am currently engaged as an AWS Cloud Consultant (AWS Java Python Developer) at The Vanguard Group in Malvern, Pennsylvania, through Randstad Digital, effective August 2026, as documented in Exhibits 97 and 98 of my petition. I intend to continue this engagement and to deliver AWS-based Java and Python systems for a firm that manages more than $9 trillion in assets. If this engagement ends, I will continue the same work with another U.S. financial institution or technology firm that needs this expertise, as I did at JPMorgan Chase for 55 months.

Near-term plans (the next one to two years). I will continue to design resilient, cost-efficient, and secure cloud platforms for U.S. financial institutions. I will apply the practices I developed at JPMorgan Chase, including cloud cost governance, observability, secure partner integrations, and faster release processes, to the platforms I support. I will also continue the applied research that connects my industry work to AI, including the optimization of backend and cloud infrastructure in large-scale financial systems, which I published in February 2026.

Longer-term plans (the next three to five years). I intend to expand my independent applied research and consulting on AI-driven optimization for public infrastructure. I will extend my TOPSIS-based and multi-objective models, which were funded through the U.S. Department of Transportation and the California State University Transportation Consortium. I will do this in collaboration with U.S. universities, transportation research centers, and public agencies, including those whose researchers have already cited my work. I will also continue to write on AI system design, building on my book, *Design Patterns for AI Agents*.

Service to the field. In every stage, I will continue to serve as a peer reviewer for leading AI venues, including NeurIPS, ICLR, and ICML, to review applications on the IEEE Senior Member panel, and to mentor engineers through my professional memberships in IEEE and BCS.

I will pursue this work only in my area of expertise. My record shows that I have done so consistently for more than five years, and nothing in my plans departs from it.

II. National Importance of the Endeavor

The integrity of the U.S. financial system and the modernization of its public infrastructure are pillars of national interest. My work serves both.

In FinTech, I build systems that millions of Americans rely on. At JPMorgan Chase, I engineered backend services for the Chase Shopping Hub, a platform serving more than 80 million customers.[1] My systems processed more than 5 million customer requests per month with 99.9% reliability. This work is critical to maintaining data integrity, secure partner integrations, and public trust in one of the nation's largest financial institutions. At Vanguard, which manages more than $9 trillion in assets for over 50 million investors, I apply the same distributed-systems discipline to AWS-based Java and Python systems.[2]

In transportation, my research, supported by the U.S. Department of Transportation through the Mineta Transportation Institute and the California State University Transportation Consortium, provides AI-driven models for infrastructure planning. This work helps public agencies optimize transportation networks, reduce traffic delays that cost the U.S. economy billions of dollars,[3] and improve safety at highway-rail crossings. It supports national initiatives such as the Infrastructure Investment and Jobs Act.

Dr. Zhenyu Wang, a Senior Research Associate at the Center for Urban Transportation Research at the University of South Florida, writes that U.S. public transportation infrastructure is aging and plagued by funding and equity gaps. He states that addressing them requires a scarce interdisciplinary skill set that bridges civil engineering and advanced AI, and that my work provides a practical solution. He further reports that my research has influenced multiple studies at his center.

Furthermore, my work in writing about and reviewing AI research helps ensure that the United States maintains its competitive edge in a technology fundamental to future economic and national security.

III. Practical Impact and Innovation (EB-1A Criteria Met)

My career is marked by original, high-impact contributions that others in my field have recognized and adopted.

A. Original contributions in transportation research. I joined the Mobility Research and Innovation Lab at California State University, Long Beach, in January 2020 as a research assistant on a project funded by the California Department of Transportation (Caltrans) and supported by U.S. Department of Transportation grants. As my supervisor Dr. Shailesh Chandra attests, I played a central role in automating complex analyses for California's freight initiatives. I built Python-based simulation and optimization models, including bi-objective and multi-objective frameworks that balance cost, safety, and operational performance in grade separation projects, and I worked on transportation data validation using AI and blockchain methods. I also developed a TOPSIS-based multi-criteria optimization framework that identifies California High-Speed Rail stations with the highest potential for economic return.

The significance of this research extends beyond California. Dr. Maxim Dulebenets, an Associate Professor at the FAMU-FSU College of Engineering, explains that he cited my study "Evaluating Financing Mechanisms and Economic Benefits to Fund Grade Separation Projects" in his own study of rail delays at level crossings in Florida. He adds that my work has shaped aspects of his team's freight network modeling and equity-focused analyses. My work has likewise been cited by researchers in Italy, Iran, and Colombia.

B. Scholarly articles. I have authored nine peer-reviewed scholarly articles, which have received more than 52 citations (h-index of 4). They include reports published by the Mineta Transportation Institute, a U.S. Department of Transportation-funded research center: "Evaluating Financing Mechanisms and Economic Benefits to Fund Grade Separation Projects" (2021, cited 12 times), "Evaluating Innovative Financing Mechanisms for the California High-Speed Rail Project" (2021, cited 7 times), "Framework for Rail Transport Inequality Assessment" (2024), and "Optimizing Multimodal Transportation Access with Social Distancing" (2022). My most recent article, on AI-driven optimization of backend and cloud infrastructure in large-scale financial systems, was published in IJARSCT in February 2026.

C. Original technical contribution: Design Patterns for AI Agents. In October 2025, I published *Design Patterns for AI Agents: A Structured Approach to Building Robust and Intelligent Systems* (ISBN 979-8268912371) as lead author. The book adapts classical software design patterns, such as Observer, Factory, Strategy, Broker, and State Machine, to agentic AI systems, with patterns for high-frequency trading, safety-critical systems, and healthcare compliance. Rahul Vishwakarma, a Fellow of the British Computer Society, IEEE Senior Member, and holder of more than 60 U.S. patents, states that he applied several of these patterns in his own professional practice. Engineering colleges have also adopted the book into their curricula.

D. Published material and media recognition. I have been quoted as a subject-matter expert in leading U.S. technology publications, including TechTarget, Dice Insights, and VKTR, and I was interviewed by TechBullion.[4] IEEE profiled me as an "IEEE Young Creator" across its official

channels, including IEEE Transmitter.[5] Dainik Bhaskar and Rajasthan Patrika, two of India's largest newspapers, also covered my knowledge-sharing work. I have also presented my applied AI research at invited industry-academia lecture series and an AICTE-approved faculty development program hosted by Anand International College of Engineering, as Sinchan Banerjee, Director of Enterprise Data Architecture at UST and an IEEE Senior Member, confirms.

IV. Why I Am Well-Positioned to Execute This Endeavor (EB-1A Criteria Met)

I am uniquely positioned to lead this endeavor because I have already risen to the top of my field, as shown by my critical roles and my selection to judge other experts. Meeting multiple criteria for this classification, including performing in a critical role for distinguished organizations,[6] confirms my sustained national and international acclaim.

A. Critical role at JPMorgan Chase & Co. (October 2021 – May 2026). For 55 consecutive months, I worked as a core backend engineer at JPMorgan Chase, first as an Associate Software Engineer and later as a Senior Software Engineer, on the Chase Shopping Hub. The platform is central to the bank's consumer strategy and serves more than 80 million customers. My responsibilities included building critical APIs from scratch and engineering the secure infrastructure beneath the platform.

Corey Graham, Executive Director and Director of Software Engineering for Consumer & Community Banking, confirms that I designed backend services, integrated external vendors for rewards redemption and settlement, configured observability pipelines, and ensured compliance with the firm's enterprise standards. He adds that I worked closely with the Level 2 engineering and reliability teams and played a hands-on role in stabilizing backend systems during a fast-moving public rollout. He also notes that I mentored engineers, reviewed code, and supported the onboarding of new developers.

Sumit Bhatnagar, a Vice President in JPMorgan Chase's Technology Division, evaluated my work independently. He states that I took full ownership of the platform's backend architecture with the autonomy of a senior technical lead. He reports that my changes cut response times from 53 to 30 milliseconds, a 43% improvement, which increased customer satisfaction by 15%. My cost-control solutions for AWS saved approximately $100,000, and my redesign of deployment processes reduced release cycles from days to hours. He also reports that the platform launched with zero downtime. He further states that I established standards for code quality and security, and that he applied my cloud cost governance approach across his own teams.

B. Critical role at The Vanguard Group (August 2026 – present). The Vanguard Group, the world's second-largest investment manager, engaged me through Randstad Digital as an AWS Cloud Consultant (AWS Java Python Developer) specifically for my expertise in AWS cloud infrastructure, Java, and Python. I apply the large-scale distributed-systems principles I developed at JPMorgan Chase to a firm managing more than $9 trillion in assets.

C. Judging the work of others. I have repeatedly been invited to evaluate the work of other experts at the highest levels of my field. I was selected as a reviewer for the NeurIPS 2026 main conference and completed peer reviews of four papers. I have also reviewed for the ICLR 2025 FM-Wild

Workshop, the NeurIPS 2025 Position Paper Track, and the ICML 2026 AIWILD Workshop.[7] These conferences are known for rigorous, expert-only review and exceptionally low acceptance rates (typically 20-25%).[8] In addition, I served on the IEEE Senior Member Application Virtual Review Panel in June 2025, evaluating whether other experienced professionals merit elevation to Senior Member. I have reviewed for international conferences and scholarly journals, and I have been invited as a judge for major U.S. hackathons, including PennApps (University of Pennsylvania), MorganHacks (Morgan State University), and DragonHacks (Drexel University).[9]

D. Membership in associations requiring outstanding achievement. I am an IEEE Senior Member, the highest grade for which a member can apply, which is held by only about 8% of IEEE's more than 460,000 members.[10] The grade requires at least ten years of professional practice, five years of significant performance in the field, and endorsement by three Senior Members or Fellows. I am also a Fellow of BCS, The Chartered Institute for IT (FBCS), the highest applicable grade of the United Kingdom's Royal Charter professional body for information technology, which has over 70,000 members in more than 150 countries.

V. Conclusion

My record of sustained national and international acclaim confirms that I am among the small percentage of experts who have risen to the very top of my field. My work is not theoretical. I have served in critical, high-stakes roles at JPMorgan Chase and The Vanguard Group, and my research has been adopted and cited by government-affiliated researchers across several countries. My status as an IEEE Senior Member and BCS Fellow, together with my selection as an expert reviewer for NeurIPS, ICLR, and ICML, shows that my expertise is acknowledged at the highest international level.

Allowing me to continue my work in the United States will provide substantial, tangible benefits to the national interest. My continued endeavor will strengthen National Economic Security by architecting resilient FinTech and cloud platforms for major institutions, such as the Chase Shopping Hub, which serves more than 80 million customers, and Vanguard's investment platforms, where my work protects data integrity and public trust in the U.S. financial system. I will also advance Intelligent Public Infrastructure, as my U.S. Department of Transportation-funded research has already led to internationally cited optimization models used in high-speed rail planning, providing AI-driven solutions to reduce congestion, improve safety, and support national initiatives such as the Infrastructure Investment and Jobs Act.

Furthermore, I will help ensure U.S. Technological Leadership by serving as an expert reviewer for premier AI conferences and by mentoring engineers through the IEEE and through my writing, helping maintain the nation's competitive edge in the critical field of Artificial Intelligence.

I respectfully submit that my extraordinary ability is essential to these vital areas of U.S. technological, financial, and infrastructural advancement. My proven capacity to lead mission-critical projects and deliver internationally recognized innovation demonstrates that I qualify for the EB-1A classification. I respectfully request that my petition be approved, allowing me to continue applying my expertise for the sustained benefit of the United States.

I declare this under penalty of perjury under the laws of the United States of America.

Sincerely,

Vivek Mishra

[1] https://www.jpmorganchase.com/ir/news/2024/chase-hosts-2024-investor-day

[2] https://about.vanguard.com/who-we-are/

[3] https://www.inrix.com/press-releases/2022-global-traffic-scorecard/

[4] https://www.techtarget.com/corporate/about-us/ ; https://www.dice.com/about-us ; https://techbullion.com/about-us/

[5] https://transmitter.ieee.org/impact-creators/

[6] https://www.jpmorganchase.com/about ; https://about.vanguard.com/who-we-are/

[7] https://neurips.cc/ ; https://iclr.cc/ ; https://icml.cc/

[8] https://medium.com/@ProgramChairsICLR2024/iclr-2024-reviewing-stats-and-reviewer-highlighting-469850afa87d ; https://www.techtarget.com/searchenterpriseai/definition/NeurIPS

[9] https://pennapps.com/ ; https://morganhacks.org/ ; https://dragonhacks.org/

[10] https://www.ieee.org/membership/senior-member.html